Ollie Liu

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EXPERIENCE

Reflection AI *Member of Technical Staff* Apr 2026 – Present

- Building reinforcement learning systems for open-weight models. Core contributor and **alignment RL captain** for **Beam**, Reflection's first open-weight model (501B-parameter sparse MoE) [tech report].
- Owned SFT and RL training of Beam's alignment expert, which was merged into the released model via multi-teacher on-policy distillation. Ran the release-candidate cycle: integrated data and RL environments from seven workstreams (instruction following, factuality, long context, agentic tool use, conversational behavior, safety, model identity), ran ablations, and coordinated trace reviews to lock the final recipe.
- Beam ranks **#1 among American open-weight models** on Arena's Alignment Index (#2 among all open models).

EDUCATION

University of Southern California *PhD in Computer Science* Advisors: Dani Yogatama and Willie Neiswanger 2026

Northwestern University *MS in Operations Research* 2021

Carnegie Mellon University *BS/MS in Machine Learning* 2019

SELECTED PUBLICATIONS

⋆ equal contribution. Full list at Google Scholar.

- **Beam Model Technical Report**. Reflection Team (**core contributor**). *Reflection AI, 2026*.
- **AION-1: Omnimodal Foundation Model for Astronomy**. Polymathic AI Collaboration, Lanusse⋆, Parker⋆, Shen⋆, **Liu**, et al. *NeurIPS 2025;* Oral, AI for Science Workshop.
- **DeLLMa: Decision Making Under Uncertainty with Large Language Models**. **Liu**⋆, Fu⋆, Yogatama, Neiswanger. *ICLR 2025,* Spotlight.
- **METAGENE-1: Metagenomic Foundation Model for Pandemic Monitoring**. **Liu**, Jaghouar, Hagemann, Wang, Wiemels, Kaufman, Neiswanger. *Preprint, 2025*.
- **Interpretable Diffusion via Information Decomposition**. Kong⋆, **Liu**⋆, Li, Yogatama, Ver Steeg. *ICLR 2024*.
- **How Does GPT-2 Compute Greater-Than? Interpreting Mathematical Abilities in a Pre-Trained Language Model**. Hanna, **Liu**, Variengien. *NeurIPS 2023*.

RESEARCH EXPERIENCE

Simons Foundation *Research Scientist* with François Lanusse 2024 – 2026

Meta FAIR *Research Intern* with Brandon Amos 2025

Microsoft Research *Research Intern* with Corby Rosset 2024

Reka AI *Research Intern* with Qi Liu and Dani Yogatama 2023

Redwood Research *Research Resident, REMIX Program* 2023

SELECTED HONORS AND SERVICE

Fellowships: Technical Innovation Fellowship, USC (2025); University Organizer Fellowship, Open Philanthropy (2024); Provost's Fellowship, USC (2022)

Grants: Nebius Research Credits Program, est. $200,000 (2025)

Reviewer: ICML, NeurIPS, ICLR, ACL Rolling Review (2024–2026)

SKILLS

Technical: Python, PyTorch, JAX, Hugging Face; large-scale LLM post-training (SFT, asynchronous RL) and evaluation

Languages: Chinese (native), English (fluent)